



SCEGGS Darlinghurst

2008

**Preliminary Course
Semester 2 Examination**

Mathematics

**Outcomes Assessed: P2 – P8
Task Weighting: 40%**

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Attempt **all** questions and show all necessary working
- Answer all questions on the pad paper provided
- Write your Student Number at the top of each page
- **Begin each question on a new page**
- Marks will be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used

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Centre Number

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Student Number

Total marks – 78

- Attempt Questions 1 – 6

Question	Calc	Comm	Reasoning	Marks
1			/3	/13
2	/5	/1		/13
3		/2	/2	/13
4	/1	/1	/8	/13
5	/3	/2	/6	/13
6	/3	/3	/5	/13
TOTAL	/12	/9	/24	/78

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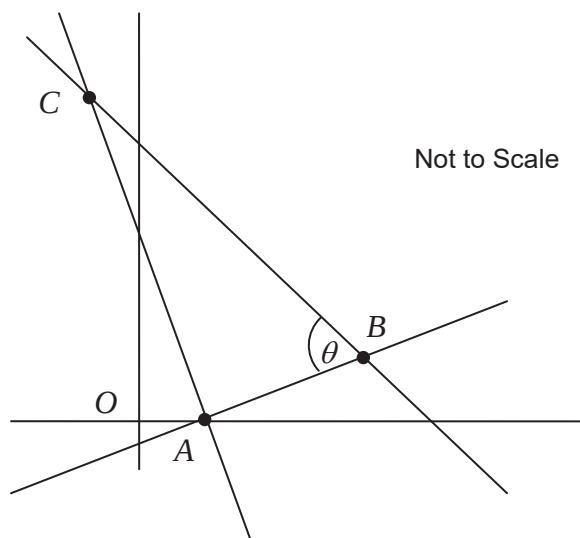
Question 1 (13 marks)

- (a) Evaluate $\frac{542}{3.17 \times 10^{15}}$ expressing your answer in scientific notation correct to 3 significant figures. **2**
- (b) Simplify $|-7| - |4|$ **1**
- (c) A car salesman buys a second hand car and then sells it at a profit of 37.5%. If the car salesman sells the car for \$30 800, what price did he pay for the car originally? **2**
- (d) Express $\frac{1}{4 - \sqrt{3}} + \frac{1}{4 + \sqrt{3}}$ in simplest form. **2**
- (e) Factorise fully:

$$8 - x^3$$
 2
- (f) Solve:
- (i) $\frac{1}{2^x} = 8$ **1**
- (ii) $|2x - 5| = 7 - 3x$ **3**

Question 2 (13 marks)

(a)



In the above diagram $A(1, 0)$, $B(4, 1)$ and $C(-1, 6)$ are points on the number plane.

Copy or trace this diagram into your writing booklet.

(i) Show that the equation of AC is $3x + y - 3 = 0$ 2

(ii) Find the length of AB 1

(iii) Show that $AB \perp AC$ 2

(iv) Find $\tan \theta$ 2

(v) On your diagram, shade the region satisfying the inequality $3x + y - 3 \leq 0$ 1

(b) Differentiate:

(i) $2x^3 - \frac{1}{x} + 5$ 1

(ii) $(3x^2 + 4)^5$ 2

(iii) $x\sqrt{1-x}$ 2

Question 3 (13 marks)

(a) Solve $\sqrt{2} \sin \theta + 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$ 2

(b) Solve $12 + 4x - x^2 > 0$ 2

(c) A ship sails from point A on a bearing of 237° for a distance of 423 kilometres to point B . The ship then turns and sails due south to point C . The bearing of A from C is 041° .

(i) Draw a diagram showing this information 1

(ii) Find the size of $\angle BAC$, to the nearest degree 1

(iii) Calculate the distance (to the nearest km) the ship must travel back to point A from C . 2

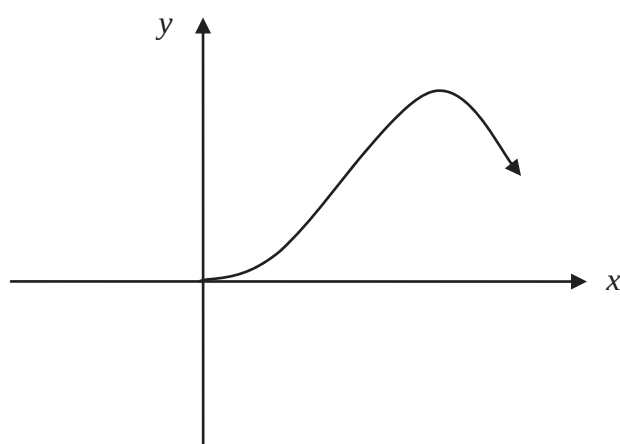
(d) If α and β are the roots of the quadratic equation $2x^2 - 6x + 3 = 0$ find:

(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$ 1

(iii) $\alpha^2 + \beta^2$ 2

(e) The diagram below shows part of the graph of an even function 1



Copy and complete the graph of the function.

Question 4 (13 marks)

(a) Find the quadratic equation whose roots are 2 and -3 1

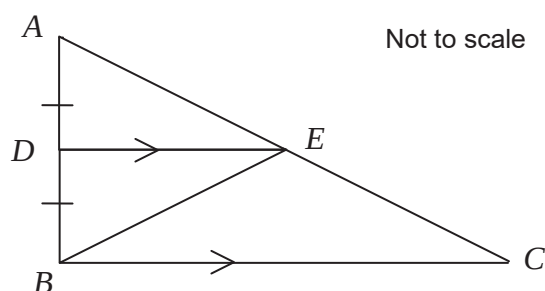
(b) Consider the function $y = \frac{1}{x} + 2$

(i) State the domain and range of the function 2

(ii) Find $\frac{dy}{dx}$ 1

(iii) Hence or otherwise explain why the gradient of $y = \frac{1}{x} + 2$ is always negative 1

(c)



The triangle ABC has a right angle at B . D is the midpoint of AB and DE is parallel to BC . Copy this diagram into your writing booklet.

(i) Prove that ADE is a right angle 1

(ii) Prove that triangle AED is congruent to triangle BED 2

(iii) Prove that $BE = EC$ 2

(d) Express $3x^2 - 7x - 2$ in the form $ax(x + 1) + bx^2 + c(x + 1)$ 3

Question 5 (13 marks)

- (a) The angles in a regular polygon are 156° each. Find the number of sides in the polygon. 2

- (b) Find the values of q for which the expression $2x^2 - x + 4q$ is positive definite 3

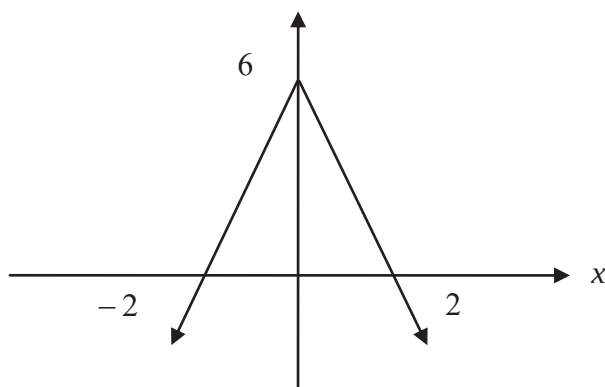
- (c) Differentiate from first principles 3
 $f(x) = 2x^2 - 6x$

- (d) Prove that: 3

$$\sin \theta (1 + \tan \theta) + \cos \theta (1 + \cot \theta) = \frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta}$$

- (e) Maud was asked to sketch the graph of $y = |6 - 3x|$ showing all important features 2

Below is her solution:



The solution is incorrect.

Explain why this is incorrect and draw a correct solution.

Question 6 (13 marks)

(a) Simplify $\tan^2 \theta (1 - \sin^2 \theta)$ **2**

(b) For the function defined by:

$$f(x) = \begin{cases} 3 - x^2 & \text{for } -2 \leq x \leq -1 \\ 2x & \text{for } -1 < x < 1 \\ x^2 + 1 & \text{for } 1 \leq x \leq 2 \end{cases}$$

(i) Evaluate: $(\alpha) f(-2)$ **1**

$(\beta) f(1)$ **1**

(ii) Sketch the graph of the function in the given domain. **3**

(c) Find the value of x for which the curve $y = (3x - 4)^3$ cuts the x -axis and find the gradient of the tangent at this point. **3**

(d) Solve $x^4 - 5x^2 - 36 = 0$ **3**

End of paper

2008 Mathematics Sem 2 Exam Solutions

Q1

SL

Reas 3

a) 1.71×10^{-13} ✓✓ (correct s.f.)

b) $7 - 4 = 3$ ✓

c) $137.5\% = \$30\ 800$ ✓

$1\% = \$224$

$100\% = \$22\ 400$ ✓

d) $\frac{1}{4 - \sqrt{3}} + \frac{1}{4 + \sqrt{3}}$

$= \frac{4 + \sqrt{3} + 4 - \sqrt{3}}{16 - 3}$ ✓

$= \frac{8}{13}$ ✓

e) $8 - x^3 = (2 - x)(4 + 2x + x^2)$ ✓✓

f) i) $2^{-x} = 2^3$

$-x = 3$

$x = -3$ ✓

ii) $|2x - 5| = 7 - 3x$

Reas 3

① $2x - 5 = 7 - 3x$ ② $-2x + 5 = 7 - 3x$

$5x = 12$

$x = 2$

$x = 2\frac{2}{5}$

✓ both solutions

check

LHS = $|2 \times 2\frac{2}{5} - 5|$

LHS = $\frac{1}{5}$

RHS = $7 - 3 \times 2\frac{2}{5}$

RHS = $-\frac{1}{5}$

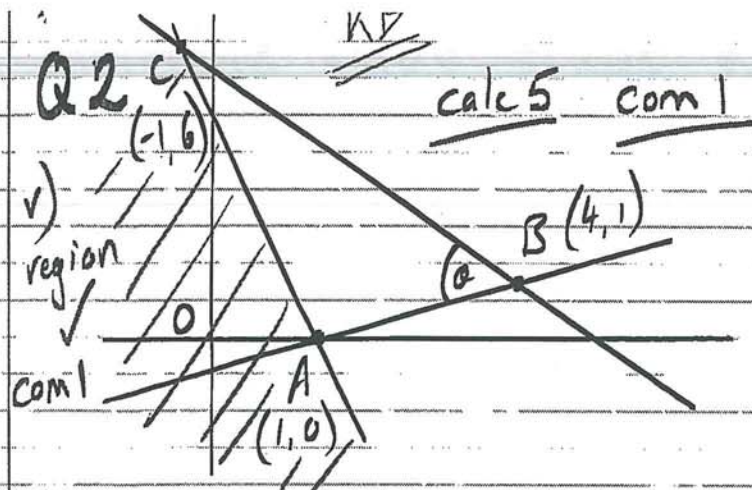
LHS = $|2 \times 2 - 5|$

RHS = $7 - 3 \times 2$

∴ only solution is $x = 2$

✓ checks

✓ conclusion



Diagrams often too small. Ruler and pencil is required.

i) $AC: m = \frac{6-0}{-1-1}$
 $= -3$ ✓

eqn: $y - 0 = -3(x - 1)$

$y = -3x + 3$ ✓

$3x + y - 3 = 0$

ii) $AB = \sqrt{(4-1)^2 + (1-0)^2}$
 $= \sqrt{9 + 1}$
 $= \sqrt{10}$ ✓

iii) $M_{AB} = \frac{1-0}{4-1}$ $M_{AC} = -3$
 $= \frac{1}{3}$ ✓

$M_{AB} \times M_{AC} = \frac{1}{3} \times -3$
 $= -1$ ✓

must explain this step.

$\therefore AB \perp AC$

iv) $AC = \sqrt{(1-(-1))^2 + (0-6)^2}$
 $= \sqrt{4 + 36}$
 $= \sqrt{40}$
 $= 2\sqrt{10}$ ✓

$\tan \theta = \frac{AC}{AB}$

$= \frac{2\sqrt{10}}{\sqrt{10}}$

$\tan \theta = 2$ ✓

must have $\tan \theta = 2$.

The angle was not requested.

Read questions carefully.

b) i) $2x^3 - x^{-1} + 5$

Calc 5

$$\frac{d}{dx} = 6x^2 + x^{-2}$$

$$\frac{d}{dx} = 6x^2 + \frac{1}{x^2} \quad \checkmark$$

ii) $\frac{d}{dx} = 5 \cdot 6x (3x^2 + 4)^4 \quad \checkmark$

Remember to differentiate

$$3x^2 + 4$$

$$\frac{d}{dx} = 30x (3x^2 + 4)^4 \quad \checkmark$$

iii) $x (1-x)^{\frac{1}{2}}$

This is a product Rule

Question.

$$u = x \quad v = (1-x)^{\frac{1}{2}}$$

$$u' = 1 \quad v' = -\frac{1}{2} (1-x)^{-\frac{1}{2}}$$

$$= \frac{-1}{2\sqrt{1-x}} \quad \checkmark$$

Very poor answers in many cases.

$$\frac{d}{dx} = (1-x)^{\frac{1}{2}} + \frac{-x}{2\sqrt{1-x}} \quad \checkmark$$

Q3

K.E.

Com 2

Reas 2

$$a) \sqrt{2} \sin \theta + 1 = 0$$

$$\sin \theta = -\frac{1}{\sqrt{2}}$$

$\sin$ -ve in Q3, Q4

acute $\angle$: $\sin \theta = \frac{1}{\sqrt{2}}$

$$\theta = 45^\circ$$

$$\therefore \theta = 180 + 45, 360 - 45$$

$$\theta = 225^\circ, 315^\circ$$

Learn this technique.
and follow the steps

$$b) 12 + 4x - x^2 > 0$$

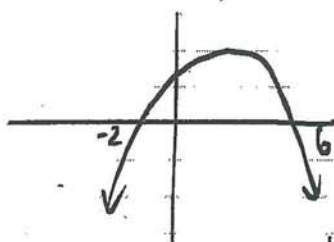
$$-(x^2 - 4x - 12) > 0$$

$$-(x-6)(x+2) > 0$$

Reas 2

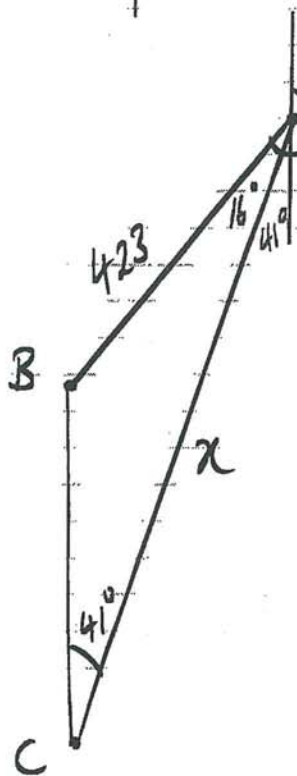
Very badly done.

Usually successful if
a graph is drawn to
solve the inequality
Factorise carefully.



$$\therefore -2 < x < 6$$

c)



$$237^\circ \text{ (i) } \checkmark$$

Com 1

(ii)

$$\angle BAC = 237 - 41 - 180$$

$$= 16^\circ \checkmark$$

$$\text{(iii) } \angle ABC = 123^\circ$$

$$\frac{x}{\sin 123^\circ} = \frac{423}{\sin 41^\circ} \checkmark$$

$$x = 541 \text{ km } \checkmark$$

Diagrams often far
too small.

Otherwise mostly well
done.

$$d) 2x^2 - 6x + 3 = 0$$

$$i) \alpha + \beta = \frac{6}{2} = 3 \quad \checkmark$$

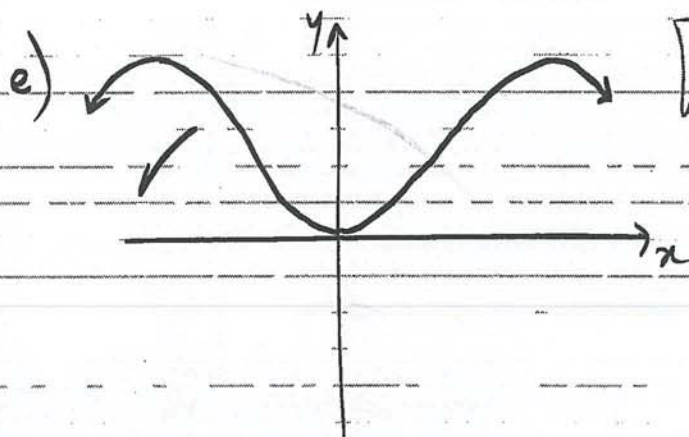
$$ii) \alpha\beta = \frac{3}{2} \quad \checkmark$$

$$iii) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \quad \checkmark$$

$$= 3^2 - 2 \times \frac{3}{2}$$

$$= 6 \quad \checkmark$$

Most students knew
this works.



Corn 1

Some really had
graphs here! They did
not really look symmetric
though that was clearly
the intention.

Q4 ^{AV} Calc 1 Com 1 Reas 8

a) $(x-2)(x+3) = 0$ ✓

well done, make sure you put $= 0$

b) $y = \frac{1}{x} + 2$

i) D: all real x , where $x \neq 0$ ✓

R: all real y , where $y \neq 2$ ✓

most were able to determine the asymptotes correctly although mixed up the x & y phrase also include all real x or all real y for domain or range not just domain: $x \neq 0$

ii) $y = x^{-1} + 2$

Calc 1

$\frac{dy}{dx} = -x^{-2}$ ✓

$= -\frac{1}{x^2}$

Well Done

iii) $-\frac{1}{x^2}$ is the gradient function

Com 1

since $x^2 > 0$

then $\frac{1}{x^2} > 0$

$\therefore -\frac{1}{x^2} < 0$

Not well explained. Try to break down the gradient function as shown.

Or use a graph of the function to explain why the gradient is always negative.

hence the gradient must always be negative

c) i) $\angle ADE = \angle ABC$ (corresponding angles)

$= 90^\circ$

are equal on parallel lines

Reas 5

Please learn your REASONS and clearly set out each step of your work.

ii) $\angle ADE = 90^\circ$ (supplementary angles)

$\therefore \angle EDB = 90^\circ$ (add to 180°)

$AD = BD$ (given) ✓

DE is common

$\therefore \triangle AED \equiv \triangle BED$ (SAS) ✓

For each statement you make you must have a reason

NEVER ASSUME something is true unless you have proved it.

iii) Prove $BE = EC$

$$\angle EBC = \angle DEB \text{ (alternate angles are equal on parallel lines)}$$

$$\angle AED = \angle ECB \text{ (corresponding angles are equal on parallel lines)}$$

$$\angle AED = \angle BED \text{ (corresponding angles in congruent triangles are equal)}$$

Some students used ratio of intercepts are equal on parallel lines which was also correct.

On the whole geometry was not well set out.

$$\therefore \angle EBC = \angle ECB$$

$$\therefore BE = EC \text{ (equal sides in isos. triangle BEC)}$$

d) $ax(x+1) + bx^2 + c(x+1)$ Reas 3

$$= ax^2 + ax + bx^2 + cx + c$$

$$= x^2(a+b) + x(a+c) + c$$

Those who had learnt their work had no problem getting to the solution.

compare $3x^2 - 7x - 2$

$$c = -2, \quad a+c = -7, \quad a+b = 3$$

$$a-2 = -7 \quad -5+b = 3$$

$$a = -5$$

$$b = 8$$

$$\therefore 3x^2 - 7x - 2$$

$$= -5x(x+1) + 8x^2 - 2(x+1)$$

Some lost 1 mark for not completing the question and writing in the form

$$ax(x+1) + bx^2 + c(x+1)$$

Q5 ^{MT} Calc 3 Com 2 Reas 6

a) int $L = 156^\circ$

ext $L = 24^\circ$ ✓

sides = $360 \div 24$

= 15 ✓

OR use $\frac{(n-2) \times 180^\circ}{n} = 156^\circ$
and solve this equation.

this part was pretty well done

b) pos. def. means

Reas 3

① concave up $a > 0$

② no real roots $\Delta < 0$

concave up since $a=2$, $a > 0$ ✓

no real roots when $\Delta < 0$

$\therefore b^2 - 4ac < 0$

$1 - 4 \times 2 \times 4q < 0$

$1 - 32q < 0$ ✓

$-32q < -1$

$q > \frac{1}{32}$ ✓

Think about the curve. It's easier to remember the facts.



concave up $a > 0$

No roots $\Delta < 0$

You can get this mark if you correctly solve your inequality.

c) $f(x) = 2x^2 - 6x$

Calc 3

$f(x+h) = 2(x+h)^2 - 6(x+h)$

$= 2x^2 + 4xh + 2h^2 - 6x - 6h$ ✓

$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 6x - 6h - 2x^2 + 6x}{h}$ ✓

$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 6h}{h}$

$= 4x - 6$ ✓

Poorly set out even though it was in the last exam a few weeks ago!

Practise until you get it right & make sure you know the correct formula!!

d) $\sin \theta (1 + \tan \theta) + \cos \theta (1 + \cot \theta) = \frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta}$ Reas 3

LHS = $\sin \theta + \frac{\sin^2 \theta}{\cos \theta} + \cos \theta + \frac{\cos^2 \theta}{\sin \theta}$ ✓

= $\frac{\sin^2 \theta \cos \theta + \sin^3 \theta + \cos^2 \theta \sin \theta + \cos^3 \theta}{\cos \theta \sin \theta}$

= $\frac{\sin^2 \theta (\cos \theta + \sin \theta) + \cos^2 \theta (\sin \theta + \cos \theta)}{\cos \theta \sin \theta}$ ✓

= $\frac{(\cos \theta + \sin \theta)(\sin^2 \theta + \cos^2 \theta)}{\cos \theta \sin \theta}$

= $\frac{\cos \theta + \sin \theta}{\cos \theta \sin \theta} = \text{RHS}$ ✓

Poorly done n

Here are some mistakes not to make next time.

$1 + \tan \theta \neq \sec \theta$ x

$1 + \tan^2 \theta = \sec^2 \theta$ ✓

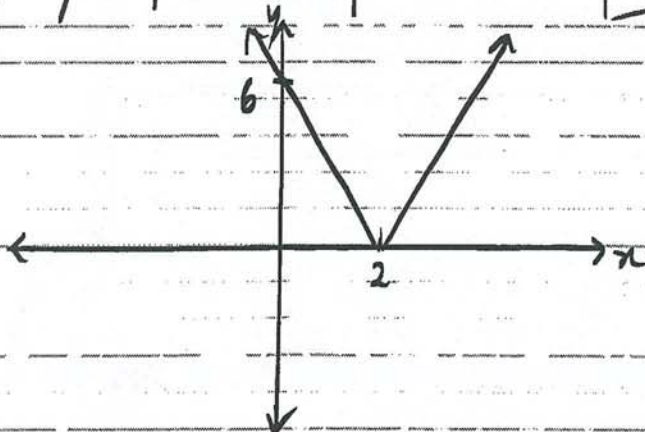
$1 + \cot \theta \neq \operatorname{cosec} \theta$ ✓

$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$ ✓

If $a^2 + b^2 = c^2$
 $a + b \neq c$ Never ✓
 $\sqrt{a^2 + b^2} \neq a + b$

e) $y = |6 - 3x|$

Com 2



the correct solution is shown above.

Maud's solution cannot be correct since: (one of the following)

- $y = |6 - 3x|$ can never go below ✓

the x-axis since the absolute value sign makes it positive.

- she has drawn $y = 6 - |3x|$

Poorly done.

Practise any questions in past papers that ask for explanations

You need to learn the correct wording.

the absolute value sign has the effect of making all the y values positive. If the curve is of the form

$y = |f(x)|$

Q 6 ^{SL} Calc 3 Com 3 Reas 5

a) $\tan^{-1} \theta (1 - \sin^2 \theta)$ Reas 2

$$= \tan^2 \theta \cdot \cos^2 \theta$$

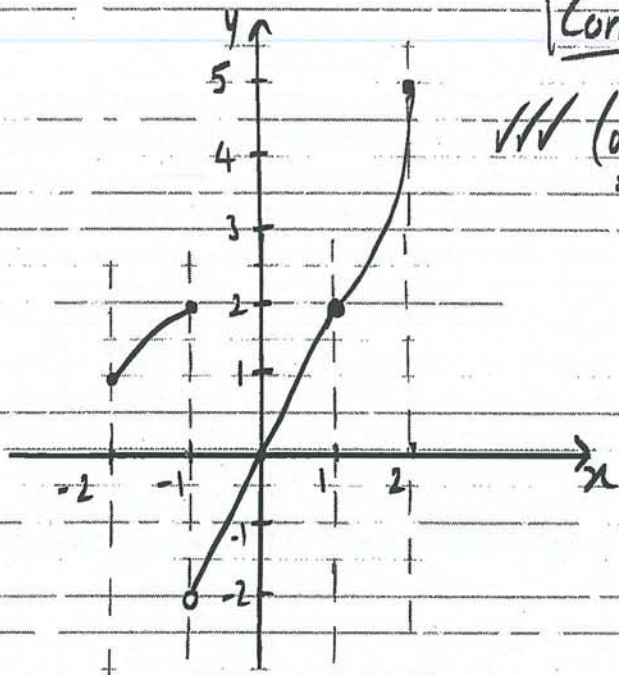
$$= \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \cos^2 \theta$$

$$= \sin^2 \theta$$

b) i) $f(-2) = 3 - 4$
 $= 1$

ii) $f(1) = 1^2 + 1$
 $= 2$

ii)



Com 3

✓✓✓ (one per section)

c) cuts x-axis when $y = 0$

Calc 3

$$0 = (3x - 4)^3$$

$$3x - 4 = 0$$

$$3x = 4$$

$$x = 1\frac{1}{3}$$

$$y = (3x - 4)^3$$

$$y' = 3 \cdot 3 (3x - 4)^2$$

$$y' = 9 (3x - 4)^2 \quad \checkmark$$

$$\text{at } (1\frac{1}{3}, 0)$$

$$m_T = 9 (3 \cdot 1\frac{1}{3} - 4)^2$$

$$= 9 (0)^2$$

$$= 0 \quad \checkmark$$

$$d) \quad x^4 - 5x^2 - 36 = 0$$

$$\text{let } u = x^2$$

$$u^2 - 5u - 36 = 0$$

$$(u - 9)(u + 4) = 0$$

$$u = 9, -4 \quad \checkmark$$

$$\therefore 9 = x^2 \quad \text{and} \quad -4 = x^2$$

$$\pm 3 = x \quad \checkmark$$

$$\text{No solns} \quad \checkmark$$